

Structural Clarification and Negative Mechanism Evidence for Kimberling's Prime-Separator Array

Research manuscript generated inside the Archivara pipeline

Primary object. Kimberling's prime-separator array T is the multiplicative greedy array whose first row is OEIS A129259 and whose full array is OEIS A129258 [OEIS Foundation Inc., 2026a,b]. Kimberling asks whether the first differences $T(1, n+1) - T(1, n)$ are bounded [Kimberling, 2026].

Manuscript status. This paper does not claim boundedness, unboundedness, or a new positive mechanism. It gives complete proofs of several exact structural properties of the recurrence, documents a digest-checked million-step computation, and incorporates the repository's late matched-control audits, all of which sharpen the evidence while leaving Kimberling's problem open.

Abstract

We study Kimberling’s prime-separator array, a multiplicative mex construction whose first row is OEIS A129259 and whose defining open problem asks whether the gap sequence $T(1, n+1) - T(1, n)$ is bounded. Our first contribution is exact and theoretical. We prove that at every step the new row and column terms are precisely the first two missing positive integers of the current restricted product set, that the two border sequences strictly interleave, that each row gap contains exactly one column term and therefore one forced axis-1 witness, that the archived `row_immediate` rule is exactly the original recurrence, that the archived `column_immediate` rule is the exact axis swap, and that every prime lies on exactly one axis. Our second contribution is computational and negative. The validated million-step run reaches record row gap 30, and the matched-control novelty-deepening corpus adds one independent checker agreement through 10^5 steps together with late record, control, and surrogate windows. These artifacts show that composite-only late record gaps persist, singleton-heavy coverage does not collapse into a bounded witness grammar, the anchor/offset lane fails as a bounded-memory mechanism, the border-prefix law fails for structural reasons, full-hypergraph leaf-fraction invariants are only near-misses, and small-modulus locking does not predict the late windows; see `results/experiments/run_1000000/record_gap_summary.json`, `results/novelty_deepening/item_026_anchor_metrics.json`, `results/novelty_deepening/item_029_hypergraph_metrics.json`, and `results/novelty_deepening/item_030_modular_metrics.json`. The resulting paper is therefore deliberately narrow: it is a structural clarification and negative-result dossier around Kimberling’s open problem, not a claimed mechanism paper.

1 Introduction

Kimberling’s prime-separator array is defined by a short recurrence with unexpectedly rigid geometry. Starting from $T(1, 1) = 1$, one chooses the next first-row term as the least positive integer missing from the current $n \times n$ multiplicative border square, then chooses the next first-column term as the least remaining miss, and finally fills the new row and column multiplicatively. The opening rows are

$$\begin{bmatrix} 1 & 2 & 4 & 7 & 9 & 13 & 15 & 18 & 23 & 25 & 29 \\ 3 & 6 & 12 & 21 & 27 & 39 & 45 & 54 & 69 & 75 & 87 \\ 5 & 10 & 20 & 35 & 45 & 65 & 75 & 90 & 115 & 125 & 145 \\ 8 & 16 & 32 & 56 & 72 & 104 & 120 & 144 & 184 & 200 & 232 \\ 11 & 22 & 44 & 77 & 99 & 143 & 165 & 198 & 253 & 275 & 319 \end{bmatrix}.$$

Every prime lies on exactly one axis, and the first-row difference sequence begins

$$1, 2, 3, 2, 4, 2, 3, 5, 2, 4, 2, 5, 4, 2, 4, 3, 2, 4, 3, 2, 2, 4, 4, 7, \dots$$

[OEIS Foundation Inc., 2026a,b]. Kimberling’s public problem asks whether these differences remain bounded [Kimberling, 2026].

The repository studied here already contains more than a term dump. It includes a validated generator, witness-carrying record-gap logs, selected full witness hypergraphs, phase-4 million-step experiments, and a later phase of matched-control novelty-deepening audits. At the same time, the repository’s own verification layer is explicit about scope: the supported contribution is not a theorem settling boundedness and not a new positive mechanism, but a tighter description of what is exact, what is reproducibly observed, and which plausible mechanism stories have now failed on the available corpus.

The paper adopts that narrower claim hierarchy. In our assessment, three comparison classes matter. First, OEIS A129258/A129259 and Kimberling’s problem page already occupy the public provenance of the object and the question [OEIS Foundation Inc., 2026a,b, Kimberling, 2026]. Second,

the closest live mathematical overlap risk is Ford’s multiplication-table and divisor-in-an-interval branch, because any explanation phrased purely in local factor coverage threatens to drift into an established restricted-product-set literature [Ford, 2006, 2008, 2011]. Third, multiplicative-basis papers form a weaker comparison class that matters mainly as a guardrail against reframing the recurrence into a different sparse-product problem [Dressler, 1970, Nathanson, 1987, Pus, 1992, Pach and Sándor, 2017]. The repo’s novelty ledger reaches exactly the same conclusion: what survives cleanly is a structural cleanup plus a negative-result evidence package, not a broad mechanism claim; see `results/literature/prior_art_gap.md` and `results/verification/novelty_report.md`.

Our contributions are therefore the following.

- C1.** We prove exact recurrence identities that were previously scattered across code and notes: the original rule picks the first two missing positive integers of the current product set; the border sequences strictly interleave; each row gap contains exactly one column term; the `row_immediate` rule is identical to the baseline; the `column_immediate` rule is the exact axis swap; and every prime lies on exactly one axis.
- C2.** We synthesize the validated baseline computation. The archived million-step run and fresh replay agree on the structural digest, reproduce the published small prefix, and extend the observed record row gap to 30.
- C3.** We integrate the repository’s late matched-control evidence. An independent checker agrees with the baseline through 10^5 steps; the shared corpus contains 5 late baseline record windows, 45 anchor-matched non-record controls, and 50 affine surrogates; and the anchor/backbone, prefix, hypergraph, and modular candidate mechanisms all fail to produce a proof-quality, horizon-independent explanation.
- C4.** We restate the open problem more sharply than the public provenance alone allows. The recurrence has a rigid local geometry, and some late-window observables are highly discriminative on the current corpus, but none of the present arguments proves boundedness or unboundedness of $T(1, n + 1) - T(1, n)$.

The rest of the paper is organized as follows. Section 2 positions the work against the provenance and overlap branches already identified in the repository. Section 3 defines the recurrence, the matched-control corpus, and the observables used later. Section 4 describes the generator, checker, novelty-deepening pipeline, and figure generation. Section 5 proves the exact structural results. Section 6 documents the deterministic computational protocol. Section 7 reports the finite-horizon findings and the negative mechanism audits. Sections 8 and 9 discuss limitations, open questions, and the appropriate claim boundary. Appendices collect extended proofs, implementation details, and supplementary tables.

2 Related Work

2.1 Public provenance

The array itself is OEIS A129258 and the first row is OEIS A129259 [OEIS Foundation Inc., 2026a,b]. Kimberling’s page “100 Conjectures and/or Problems” records the bounded-difference question explicitly [Kimberling, 2026]. Accordingly, we do not treat reconstruction of the array, reproduction of the first-row prefix, or restatement of the prime-separation phenomenon as novel. The differentiated contribution of the present paper must therefore come from formal consequences of the recurrence or from a stronger validated evidence package than the public provenance provides.

2.2 Closest overlap risk: restricted product sets

In our assessment, the most important overlap branch is Ford’s work on multiplication tables and divisor intervals [Ford, 2006, 2008, 2011]. The reason is structural rather than terminological: at step n , the recurrence asks which integers near the frontier are represented in the restricted product set $\mathcal{R}_n\mathcal{C}_n$. Any proposed explanation of long row gaps that appeals only to local factor coverage, divisor availability, or distinct-product density risks becoming a reformulation of an existing restricted-product-set question rather than a genuinely T -specific theorem. This is an author judgment, not a theorem of Ford’s papers; it is also the explicit novelty boundary recorded in `results/literature/prior_art_gap.md` and `results/verification/novelty_report.md`.

2.3 More distant multiplicative-basis comparisons

The multiplicative-basis literature provides a second comparison class [Dressler, 1970, Nathanson, 1987, Pus, 1992, Pach and Sándor, 2017]. These papers study sparse sets whose products represent integers, but they do not address Kimberling’s coupled mex dynamics. They are relevant here mainly as a warning against overreframing: if one drops the endogenous update order and treats the border sets only as a sparse multiplicative basis, the discussion moves away from Kimberling’s recurrence and toward a different established topic.

2.4 How this paper differs

Relative to the public provenance, this paper adds complete proofs of exact recurrence properties, a digest-checked computation package through one million steps, and a matched-control negative-result audit. Relative to Ford-style local coverage, the paper is narrower: it does not claim a new asymptotic theory of restricted multiplication tables, and it does not claim a T -specific positive mechanism that separates cleanly from generic local coverage. Relative to the multiplicative-basis branch, the contribution remains tied to the actual recurrence rather than to an abstract basis problem.

3 Background and Preliminaries

3.1 Definitions

Let

$$r_n := T(1, n), \quad c_n := T(n, 1)$$

for $n \geq 1$, so $r_1 = c_1 = 1$. Define the finite border sets

$$\mathcal{R}_n := \{r_1, \dots, r_n\}, \quad \mathcal{C}_n := \{c_1, \dots, c_n\},$$

and their restricted product set

$$\mathcal{P}_n := \mathcal{R}_n\mathcal{C}_n := \{xy : x \in \mathcal{R}_n, y \in \mathcal{C}_n\}.$$

The recurrence is

$$r_{n+1} = \text{mex}(\mathcal{P}_n), \quad c_{n+1} = \text{mex}(\mathcal{P}_n \cup \{r_{n+1}\}),$$

with $T(m, n) = c_m r_n$ for all $m, n \geq 1$.

Definition 3.1 (Row gap, offset, and row-gap window). For $n \geq 1$, define the *row gap*

$$g_n := r_{n+1} - r_n,$$

the *offset*

$$\delta_n := c_n - r_n,$$

and the open *row-gap window*

$$W_n := \{r_n + 1, \dots, r_{n+1} - 1\}.$$

If $g_n > 1$, then W_n is nonempty and contains the skipped values between two consecutive row terms.

Definition 3.2 (Anchor and left/right depths). When $g_n > 1$, the *anchor* of the row-gap window is the previous column term c_n . Define

$$\ell_n := c_n - r_n - 1, \quad u_n := r_{n+1} - c_n - 1.$$

Thus ℓ_n and u_n are respectively the numbers of skipped values strictly to the left and right of the anchor inside W_n .

Definition 3.3 (Witness). Fix n and a skipped value $v \in W_n$. A *witness* for v at step n is a factor pair $(x, y) \in \mathcal{R}_n \times \mathcal{C}_n$ such that $xy = v$. The baseline corpus stores one chosen witness and its multiplicity for each skipped value in each record interval; selected late gaps and matched controls also store the full admissible witness-pair set.

Definition 3.4 (Record gap). A *record gap* is an index n such that

$$g_n > \max_{1 \leq k < n} g_k.$$

3.2 Matched-control corpus

The novelty-deepening phase builds a single shared corpus from the baseline million-step run; see `results/novelty_deepening/shared_corpus.json`. The corpus contains:

- the five baseline record windows with $g_n \geq 20$,
- forty-five anchor-matched non-record control windows drawn from the same late trajectory,
- fifty affine size-matched surrogate windows.

For every such window, the corpus stores the anchor position, left/right depths, prime/composite counts, full witness-pair data, and several projected graph summaries. The point of this corpus is not to prove boundedness; it is to test whether candidate mechanism observables distinguish true record windows from comparable windows that are not records.

3.3 Artifact map used in this paper

The manuscript relies on the following repository artifacts.

- Generator and tests: `scripts/prime_separator.py`, `scripts/prime_separator_variants.py`, `tests/test_prime_separator.py`.

- Independent checker and matched-control pipeline: `scripts/novelty_deepening.py`, `tests/test_novelty_deepening.py`, `results/novelty_deepening/checker_agreement.json`, and `results/novelty_deepening/shared_corpus.json`.
- Baseline million-step contract and replay target: `results/experiments/run_1000000/contract.json`.
- Witness summaries: `results/experiments/run_1000000/record_gap_summary.json`.
- Full witness hypergraphs for late baseline gaps: `results/analysis/full_witness_hypergraphs_gap21_25_28_30.json`.
- Negative-result audit artifacts: `results/novelty_deepening/item_026_anchor_backbone.md` through `results/novelty_deepening/item_030_modular_locking.md`.
- Narrative verification documents: `results/verification/novelty_report.md`, `results/verification/citation_audit.md`, and `results/verification/verification_summary.md`.

Whenever a quantitative statement comes from computation rather than from the external literature, we cite the relevant local artifact directly.

4 Method

4.1 Validated generator

The baseline implementation `scripts/prime_separator.py` stores the current border sequences, maintains an extendable coverage array for $\mathcal{P}_n$, and updates that coverage incrementally when a new row or column term is appended. Prefix validation is built into the emitted contract: the script checks the given 5×11 table, the first-row prefix, and the first-column prefix against the problem statement. The contract digest stored in `contract.json` is a SHA-256 hash of the border sequences and record-gap locations, so exact replay can be checked mechanically.

4.2 Pseudocode

Algorithm 1. Baseline border generator

1. Initialize $r_1 = c_1 = 1$, $\mathcal{R}_1 = \mathcal{C}_1 = \{1\}$, and coverage $\mathcal{P}_1 = \{1\}$.
2. For $n = 1, 2, \dots, N - 1$:
 - 2.1. Set r_{n+1} to the least positive integer not in $\mathcal{P}_n$.
 - 2.2. Set c_{n+1} to the least positive integer not in $\mathcal{P}_n$ and different from r_{n+1} .
 - 2.3. If $g_n = r_{n+1} - r_n$ is a new record, enumerate each integer v with $r_n < v < r_{n+1}$ and store:
 - 2.3.1. prime/composite status,
 - 2.3.2. one witness $(x, y) \in \mathcal{R}_n \times \mathcal{C}_n$ with $xy = v$,
 - 2.3.3. witness multiplicity inside $\mathcal{R}_n \times \mathcal{C}_n$.
 - 2.4. Append r_{n+1} to $\mathcal{R}_n$ and mark all products $r_{n+1}c$ with $c \in \mathcal{C}_n$.
 - 2.5. Append c_{n+1} to $\mathcal{C}_n$ and mark all products rc_{n+1} with $r \in \mathcal{R}_{n+1}$.
3. Emit the border sequences, record-gap corpus, prefix checks, structural digest, and resource counters.

4.3 Complexity

Let

$$W(L) := \#\{(i, j) : r_i c_j \leq L\}$$

be the number of admissible border-factor incidences up to a coverage limit L . The implementation runs in time proportional to the total number of marked incidences plus linear candidate scans for the successive mex calls. Consequently the cost through step N is

$$O(W(L_N) + L_N),$$

where L_N is the largest internal coverage-array index reached during the run. This is an implementation bound rather than an asymptotic theorem about the sequence itself.

4.4 Independent checker and matched-control pipeline

The late phase of the project adds `scripts/novelty_deepening.py`. It performs five tasks relevant to the final paper.

- (i) It runs a separately written checker and compares its border sequences with the main generator on a substantial prefix.
- (ii) It extracts late baseline record windows and matched non-record windows from the baseline million-step run.
- (iii) It constructs affine surrogate windows of the same sizes.
- (iv) It computes anchor, offset, recent-prefix, and full-witness graph observables on all windows.
- (v) It emits summary metrics for the anchor/backbone, schedule, border-prefix, hypergraph, and modular candidate mechanisms.

This pipeline is deliberately bounded. It does not search for larger horizons or widen the hypothesis family beyond the three lanes already recorded in `results/swarm/hypotheses.json`.

4.5 Figure pipeline

All figures in this paper are programmatically generated by `scripts/make_paper_figures.py`. The script emits TSV data files and standalone TikZ/PGFPlots sources under `figures/`, compiles them to PDF with `pdflatex`, and renders 600-DPI PNG copies with `mutool draw`. No hand-edited graphics were used.

5 Theoretical Results and Proofs

5.1 Second-mex geometry

Lemma 5.1 (Frontier interval characterization). *For every $n \geq 1$,*

$$[1, r_{n+1} - 1] \subseteq \mathcal{P}_n$$

and

$$\mathcal{P}_n \cap [1, c_{n+1} - 1] = [1, c_{n+1} - 1] \setminus \{r_{n+1}\}.$$

Proof. By definition, $r_{n+1} = \text{mex}(\mathcal{P}_n)$. This means exactly that every positive integer less than r_{n+1} belongs to $\mathcal{P}_n$, which proves the first inclusion.

Now consider the second claim. By definition, c_{n+1} is the least positive integer that is neither in $\mathcal{P}_n$ nor equal to r_{n+1} . Since $r_{n+1} \notin \mathcal{P}_n$, this means the positive integers missing from $\mathcal{P}_n$ and strictly smaller than c_{n+1} consist of the single value r_{n+1} . Equivalently,

$$\mathcal{P}_n \cap [1, c_{n+1} - 1] = [1, c_{n+1} - 1] \setminus \{r_{n+1}\}.$$

□

Corollary 5.2 (Immediate row insertion fills the only hole below c_{n+1}). *For every $n \geq 1$,*

$$\mathcal{R}_{n+1}\mathcal{C}_n \cap [1, c_{n+1} - 1] = [1, c_{n+1} - 1].$$

Proof. By Lemma 5.1, the only missing integer below c_{n+1} in $\mathcal{P}_n = \mathcal{R}_n\mathcal{C}_n$ is r_{n+1} . After adjoining r_{n+1} to the row border, that missing value becomes represented as

$$r_{n+1} = r_{n+1} \cdot 1,$$

because $1 = c_1 \in \mathcal{C}_n$. All values already represented in $\mathcal{P}_n$ remain represented in $\mathcal{R}_{n+1}\mathcal{C}_n$. Therefore every positive integer below c_{n+1} lies in $\mathcal{R}_{n+1}\mathcal{C}_n$. □

5.2 Exact update-rule identities

Proposition 5.3 (The `row_immediate` rule is exactly the original recurrence). *Suppose one modifies the update order as follows: after choosing $r_{n+1} = \text{mex}(\mathcal{P}_n)$, immediately adjoin r_{n+1} to the row border and only then choose the next column term as the least positive integer missing from $\mathcal{R}_{n+1}\mathcal{C}_n$. The resulting row and column sequences are identical to (r_n) and (c_n) .*

Proof. Fix n . By Corollary 5.2, every positive integer less than c_{n+1} belongs to $\mathcal{R}_{n+1}\mathcal{C}_n$. Also, $c_{n+1} \notin \mathcal{R}_{n+1}\mathcal{C}_n$, because if c_{n+1} belonged to $\mathcal{R}_{n+1}\mathcal{C}_n$, then either it already belonged to $\mathcal{P}_n$ or it would equal $r_{n+1} \cdot 1$; the first possibility contradicts the definition of c_{n+1} , and the second is impossible because $c_{n+1} \neq r_{n+1}$. Therefore c_{n+1} is precisely the least missing positive integer of $\mathcal{R}_{n+1}\mathcal{C}_n$.

This shows that the modified rule chooses exactly the same c_{n+1} as the original rule, once the same r_{n+1} has been chosen. Since both constructions start from the same initial data $(r_1, c_1) = (1, 1)$, induction on n proves that all subsequent row and column terms coincide. □

Proposition 5.4 (Strict interleaving). *For every $n \geq 2$,*

$$r_n < c_n < r_{n+1}.$$

Consequently,

$$1 = r_1 = c_1 < r_2 < c_2 < r_3 < c_3 < \dots$$

Proof. Fix $n \geq 2$. The term r_n is the first missing positive integer of $\mathcal{P}_{n-1}$, and c_n is the second missing positive integer of the same set. Therefore $r_n < c_n$.

To prove $c_n < r_{n+1}$, observe first that by Proposition 5.3, adjoining r_n immediately does not change the next column choice. Hence c_n is already represented in $\mathcal{R}_n\mathcal{C}_{n-1}$ and therefore also in $\mathcal{P}_n = \mathcal{R}_n\mathcal{C}_n$. Since $r_{n+1} = \text{mex}(\mathcal{P}_n)$, every positive integer smaller than r_{n+1} belongs to $\mathcal{P}_n$. In particular, because $c_n \in \mathcal{P}_n$, the $\text{mex } r_{n+1}$ cannot be less than or equal to c_n . Thus $c_n < r_{n+1}$.

The displayed chain follows by applying the same argument successively for $n = 2, 3, \dots$ □

Corollary 5.5 (Unique column term in each row gap). *For every $n \geq 2$, the open interval (r_n, r_{n+1}) contains exactly one column term, namely c_n .*

Proof. By Proposition 5.4, one has $r_n < c_n < r_{n+1}$. Thus c_n lies in the interval.

Now let $m < n$. Then the interleaving chain gives $c_m < r_n$, so $c_m \notin (r_n, r_{n+1})$. Let $m > n$. Then $c_m > c_n$ and, again by interleaving, $c_m > r_{n+1}$, so $c_m \notin (r_n, r_{n+1})$. Therefore c_n is the only column term in that interval. $\square$

Corollary 5.6 (Forced unit witness). *For every $n \geq 2$, the skipped interval (r_n, r_{n+1}) contains exactly one value with a forced witness of the form $1 \cdot c$, namely c_n .*

Proof. By Corollary 5.5, the only column term in the interval is c_n . Since $1 = r_1 \in \mathcal{R}_n$, the product $1 \cdot c_n$ represents c_n inside $\mathcal{P}_n$. No other value in the interval can have the form $1 \cdot c_m$, because that would require a second column term c_m in the same interval, which Corollary 5.5 forbids. $\square$

Corollary 5.7 (Anchor decomposition). *For every $n \geq 2$ with $g_n > 1$, the row-gap window W_n decomposes as*

$$W_n = \{r_n + 1, \dots, c_n - 1\} \cup \{c_n\} \cup \{c_n + 1, \dots, r_{n+1} - 1\}.$$

Equivalently,

$$\ell_n = \delta_n - 1, \quad u_n = g_n - \delta_n - 1,$$

and therefore

$$g_n = \ell_n + u_n + 1.$$

Proof. Corollary 5.5 states that c_n is the unique column term in (r_n, r_{n+1}) . The open interval therefore splits uniquely at c_n into the skipped values to the left of the anchor, the anchor itself, and the skipped values to its right. Counting integers gives

$$\ell_n = (c_n - 1) - (r_n + 1) + 1 = c_n - r_n - 1 = \delta_n - 1$$

and

$$u_n = (r_{n+1} - 1) - (c_n + 1) + 1 = r_{n+1} - c_n - 1 = g_n - \delta_n - 1.$$

Adding the three pieces of the decomposition yields $g_n = \ell_n + u_n + 1$. $\square$

Proposition 5.8 (The `column_immediate` rule is the axis swap). *Consider the update rule that first chooses the least missing positive integer as the new column term, immediately adjoins it to the column border, and then chooses the next row term from the updated product set. If the resulting sequences are denoted (r'_n, c'_n) , then for all $n \geq 1$,*

$$r'_n = c_n, \quad c'_n = r_n.$$

Proof. We argue by induction on n . The initial values satisfy $r'_1 = c'_1 = 1 = c_1 = r_1$.

Assume $r'_k = c_k$ and $c'_k = r_k$ for all $k \leq n$. Then the product set of the swapped process at step n is

$$\mathcal{R}'_n \mathcal{C}'_n = \{r'_i c'_j : 1 \leq i, j \leq n\} = \{c_i r_j : 1 \leq i, j \leq n\} = \mathcal{R}_n \mathcal{C}_n = \mathcal{P}_n.$$

Therefore the first missing positive integer of the swapped process is exactly r_{n+1} , but under the swapped naming convention this first choice is the new column term. Hence

$$c'_{n+1} = r_{n+1}.$$

After adjoining this term to the swapped column border, the updated product set is

$$\mathcal{R}'_n(\mathcal{C}'_n \cup \{c'_{n+1}\}) = \mathcal{C}_n(\mathcal{R}_n \cup \{r_{n+1}\}),$$

which is just $\mathcal{R}_{n+1}\mathcal{C}_n$ with the factors reversed. By Proposition 5.3, the least missing positive integer of this updated set is c_{n+1} . Under the swapped naming convention this second choice is the new row term, so

$$r'_{n+1} = c_{n+1}.$$

This completes the induction. □

5.3 Prime separation

Proposition 5.9 (Every prime lies on exactly one axis). *Let p be prime. Then there exists exactly one index $n \geq 2$ such that either $p = r_n$ or $p = c_n$.*

Proof. Because r_n is strictly increasing and unbounded, so is c_n by Proposition 5.4. Hence there exists n with $p < c_{n+1}$.

Apply Lemma 5.1 at step n . Either $p = r_{n+1}$, in which case p is a row term, or $p \in \mathcal{P}_n$. Suppose $p \in \mathcal{P}_n$. Then there exist $x \in \mathcal{R}_n$ and $y \in \mathcal{C}_n$ such that $xy = p$. Since p is prime and x, y are positive integers, one of the factors must be 1. If $x = 1$, then $p = y \in \mathcal{C}_n$. If $y = 1$, then $p = x \in \mathcal{R}_n$. Thus in all cases the prime p appears on at least one axis.

It remains to prove uniqueness. By Proposition 5.4, the row and column sequences are strictly increasing and disjoint except at the common initial value 1. Since $p > 1$, it cannot be both a row term and a column term. Therefore p lies on exactly one axis. □

Remark 5.10. Proposition 5.9 recovers a public OEIS observation from the recurrence itself. The novelty here is the proof, not the fact that prime separation occurs.

5.4 What the theory does not yet imply

The preceding results settle several exact structural questions, but they do not resolve boundedness of (g_n) . In particular:

- strict interleaving gives only $\delta_n < g_n$;
- the forced unit witness identifies one covered point in each row gap, not the entire interval;
- the anchor decomposition rewrites each row gap in terms of δ_n and u_n , but does not bound either quantity globally;
- the variant identities prove that two nearby update-order modifications are symmetries rather than robustness tests, but they do not create new asymptotic information.

Thus the open problem still requires either a horizon-independent upper-bound invariant or a construction of arbitrarily long fully covered frontier intervals. Neither is presently known.

6 Experimental Setup

6.1 Deterministic protocol

All experiments are deterministic. There are no random seeds, no train/test splits in the stochastic sense, and no probabilistic hyperparameters. Every plotted mathematical quantity is therefore an exact value of the recurrence or of a deterministic statistic computed from the stored windows. Confidence intervals are degenerate for the mathematical observables and are omitted. The only nondegenerate intervals reported in this paper are runtime confidence intervals obtained from three repeated million-step runs in `results/experiments/runtime_repeats_1000000.json`.

6.2 Commands and artifacts

The primary archived commands are recorded in `results/experiments/run_1000000/experiment_note.md` and `results/experiments/variant_comparison.md`. The baseline million-step run is

```
python3 scripts/prime_separator.py -steps 1000000 -out-dir results/experiments/run_1000000
```

The late checker and shared-corpus build are exercised by `tests/test_novelty_deepening.py` and are implemented in `scripts/novelty_deepening.py`. The checker agreement artifact reports exact row and column agreement through 10^5 steps.

6.3 Hardware and software

The archived performance JSON stores runtime and maximum resident set size, and the repeat artifact stores repeated timing summaries. A fresh replay during manuscript preparation ran on the current Archivara environment:

- platform recorded by Python: `Linux-4.4.0-x86_64-with-glibc2.36`,
- Python version: `3.10.17`,
- logical memory visible to the process: `1055933872 KB`,
- TeX toolchain for figures and manuscript: `pdflatex`, `bibtex`, and `mutool`.

The fresh replay reproduced the archived structural digest

```
3680d24d2073239978438eaf02f8f05a1ed81da4e52927909c4dc47c0d0291e3
```

exactly.

6.4 Validation and tests

The unit test suites verify:

- (i) the given 5×11 prefix exactly;
- (ii) repeatability of the baseline structural digest over two independent 30,000-step runs;
- (iii) detectability of a genuinely wrong update order;
- (iv) exact equality of `row_immediate` with the baseline on a 200-step regression window;
- (v) exact axis-swap behavior of `column_immediate` on the same window;
- (vi) agreement of the separately written checker with the baseline on a 200-step test window;
- (vii) construction of a shared corpus containing record and control windows.

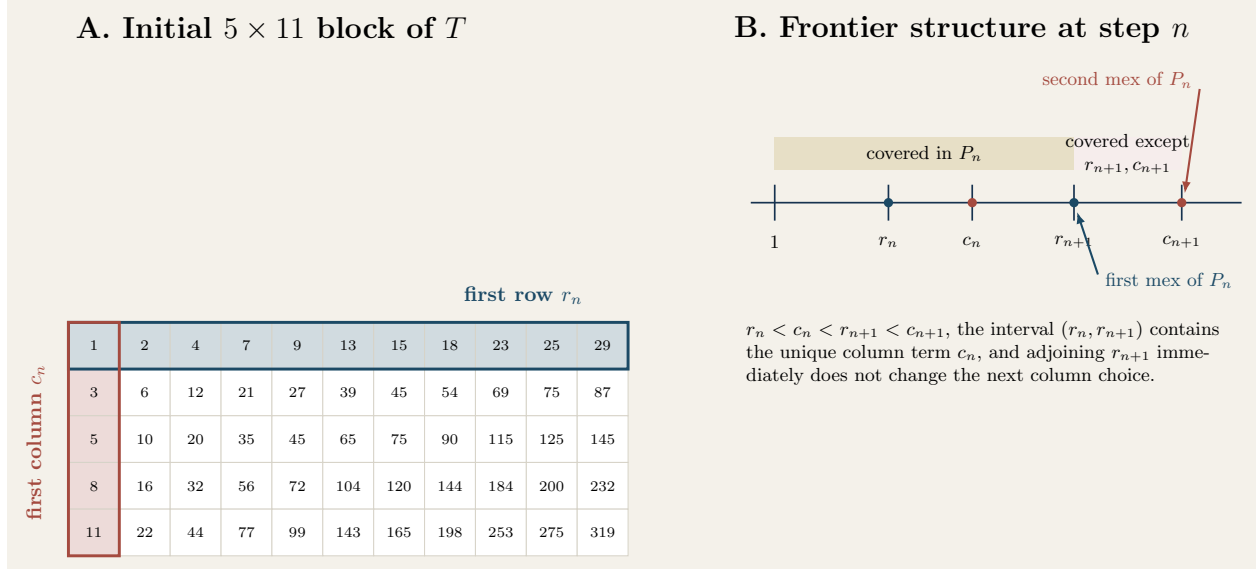


Figure 1: **Construction and frontier geometry.** Left: the initial 5×11 block of Kimberling’s array. The first row and first column determine the whole array multiplicatively. Right: the step- n frontier structure. The set $\mathcal{P}_n = \mathcal{R}_n \mathcal{C}_n$ contains every positive integer below c_{n+1} except the first missing value r_{n+1} ; after adjoining r_{n+1} , that single hole is filled by the trivial representation $r_{n+1} = r_{n+1} \cdot 1$, so the next missing value remains c_{n+1} . Proposition 5.4 and Corollary 5.7 then force $r_n < c_n < r_{n+1}$ and identify the previous column term as the unique anchor inside the row gap.

7 Results

7.1 Exact structural consequences

Figure 1 summarizes the recurrence geometry proved in Section 5. The left panel shows the initial block of the array; the right panel shows the frontier picture from Lemma 5.1, Proposition 5.4, and Corollary 5.7. The theory yields three immediate empirical consequences, all confirmed by the archived artifacts:

- (a) every row gap contains exactly one column term and therefore one forced axis-1 witness;
- (b) the anchor splits each row gap into left and right depths determined by (g_n, δ_n) ;
- (c) the archived `row_immediate` and `column_immediate` runs are identities rather than genuine ablations.

7.2 Offsets, interleaving, and the observed frontier scale

Figure 2 visualizes the exact interleaving and the offset process $\delta_n = c_n - r_n$. The interleaving itself is theorem-level. The measured quantity is the size of the offset, which never exceeds 24 on the baseline million-step run; this value occurs at step 464,334 according to direct scans of `results/experiments/run_1000000/row_terms.json` and `results/experiments/run_1000000/column_terms.json`. Corollary 5.7 predicts that the left anchor depth is $\ell_n = \delta_n - 1$, so the observed smallness of the offset explains why the previous column term sits near the left edge of many large gaps. It does *not* prove that future gaps are bounded, because $u_n = r_{n+1} - c_n - 1$ may still grow.

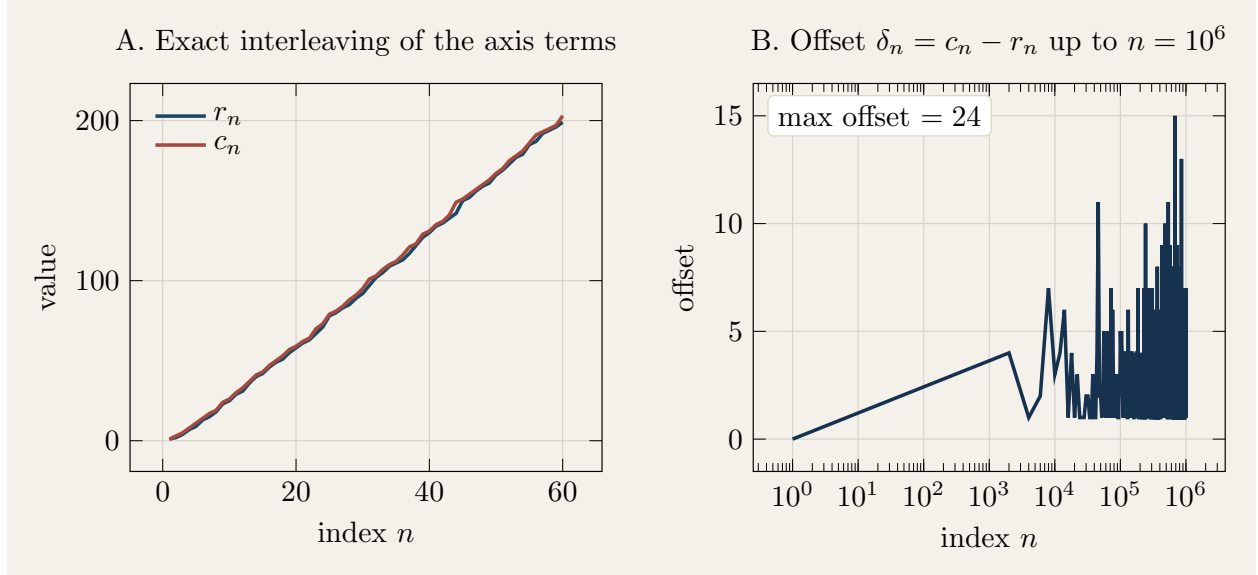


Figure 2: **Interleaving and offsets.** Left: the first 60 row and column border terms, showing the exact strict alternation proved in Proposition 5.4. Right: the sampled offset process $\delta_n = c_n - r_n$ up to $n = 10^6$. The offset stays small on the explored horizon and never exceeds 24, but the current theory does not convert this finite observation into a global theorem about the gap sequence.

7.3 Baseline record-gap trajectory

The baseline million-step run stores 17 record gaps and reaches a largest gap of 30; these numbers are exact and appear in `results/experiments/run_1000000/contract.json`. The full trajectory is listed in Table 1 and plotted in Figure 3. The three late records beyond the earlier 30,000-step horizon are

$$25 \text{ at step } 92,320, \quad 28 \text{ at step } 247,399, \quad 30 \text{ at step } 729,353.$$

The step gaps between successive records grow irregularly rather than periodically.

7.4 Witness composition and full hypergraphs

Figure 4 uses `results/experiments/run_1000000/record_gap_summary.json` to show two persistent trends across the baseline record gaps:

- (i) singleton witnesses remain common but do not converge to a rigid, tiny-factor certificate;
- (ii) the balanced-factor share rises substantially on the late record gaps, while the tiny-factor share declines.

For the late records 19, 20, 21, 25, 28, 30, the singleton shares are

$$\frac{15}{18}, \frac{17}{19}, \frac{16}{20}, \frac{18}{24}, \frac{18}{27}, \frac{19}{29},$$

whereas the balanced-factor shares rise from $1/18$ to $13/29$. Several late record gaps are composite-only: gaps 20, 21, 28, and 30 contain no skipped primes at all. These facts are exactly the ones that the repository’s negative-result narrative treats as load-bearing; see `results/experiments/run_1000000/record_gap_summary.json` and `results/final_status_note.md`.

Table 1: Baseline record gaps through 10^6 steps, from `results/experiments/run_1000000/contract.json`.

record index	step n	previous r_n	gap g_n
1	1	1	1
2	2	2	2
3	3	4	3
4	5	9	4
5	8	18	5
6	24	71	7
7	44	142	8
8	95	332	11
9	720	2923	12
10	866	3568	13
11	2063	8867	17
12	8475	38630	19
13	27676	130699	20
14	29373	139039	21
15	92320	451018	25
16	247399	1241893	28
17	729353	3760927	30

The selected full witness exports in `results/analysis/full_witness_hypergraphs_gap21_25_28_30.json` sharpen this negative conclusion rather than rescuing it. Figure 5 shows that the late gaps remain sparse even after complete witness enumeration. The most informative case is gap 30:

- skipped values: 29,
- total admissible witness pairs: 43,
- distinct row factors used: 39,
- distinct column factors used: 41,
- multiplicity histogram: 19 singletons, 8 doubletons, 1 tripleton, 1 quintuplet.

Thus the full hypergraph layer validates multiplicities and weak factor reuse, but it still does not reveal a compressed interval-level certificate.

7.5 Late matched-control audits

The strongest new evidence beyond the original manuscript comes from the novelty-deepening package. Table 2 summarizes the five audited lanes using `results/novelty_deepening/item_026_anchor_metrics.json` through `results/novelty_deepening/item_030_modular_metrics.json`.

Several points in Table 2 are worth making explicit.

First, the independent checker closes one earlier verification gap. The repository now contains a separately written implementation that agrees with the baseline through 10^5 steps, so the recurrence order is no longer justified only by repeated self-comparison.

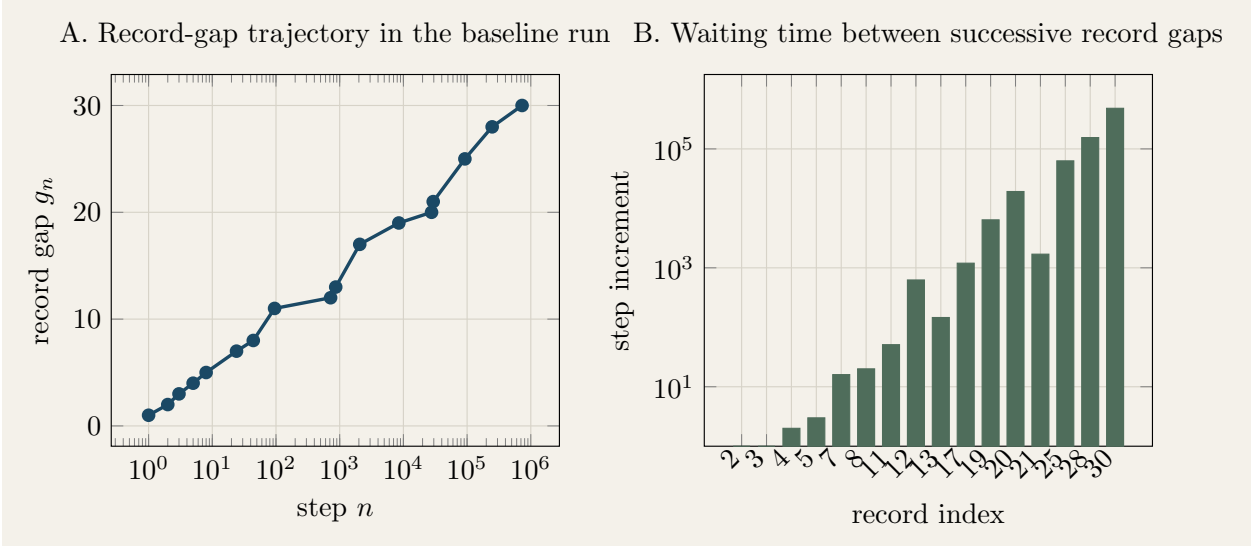


Figure 3: **Record-gap growth and waiting times.** Left: the exact record-gap trajectory from the baseline million-step run. Right: the step increments between successive records. The record process clearly continues beyond the early 30,000-step horizon, but finite-horizon growth alone does not distinguish eventual boundedness from eventual divergence.

Second, the anchor/backbone audit is informative precisely because it *fails*. The anchor itself is theorem-level by Corollary 5.5, and the left/right depths are explicit functions of (g_n, δ_n) by Corollary 5.7. On the shared corpus, that geometry strongly distinguishes true record windows from matched controls. But the late held-out records require correction pairs outside the training support, so the apparent rule does not stabilize into a bounded alphabet. This is negative evidence against a compact memory mechanism, not a positive explanation.

Third, the schedule audit turns the old “variant” narrative into a consistency result. Proposition 5.3 and Proposition 5.8 already prove the exact identities globally. The audit in `results/novelty_deepening/item_027_schedule_network.md` checks the same fact on 1000 late stored states and shows that the only schedule-independent predictive observable with strong AUROC is the right anchor depth u_n already isolated by Corollary 5.7. No new independent mechanism appears.

Fourth, the border-prefix obstruction is partly theorem and partly measurement. The theorem-level content is that the anchor value c_n can only be represented as $1 \cdot c_n$ inside the row gap. Therefore any witness rule restricted to only recent row-border terms must reach all the way back to the oldest row term. The measured depths in Table 2 show that this obstruction is not a corner case: it saturates all five late baseline record windows.

Fifth, the hypergraph and modular lanes remain negative. The full witness graphs are forests everywhere in the audited corpus, so raw cycle rank is true but useless as a discriminator. The best canonical leaf-fraction statistic is reasonably stable across canonicalizations and rejects the affine surrogates, but it still overlaps too heavily with matched non-record controls to justify a promoted invariant. The modular audit performs worse still: the best small modulus is barely above chance and cannot beat the anchor comparator.

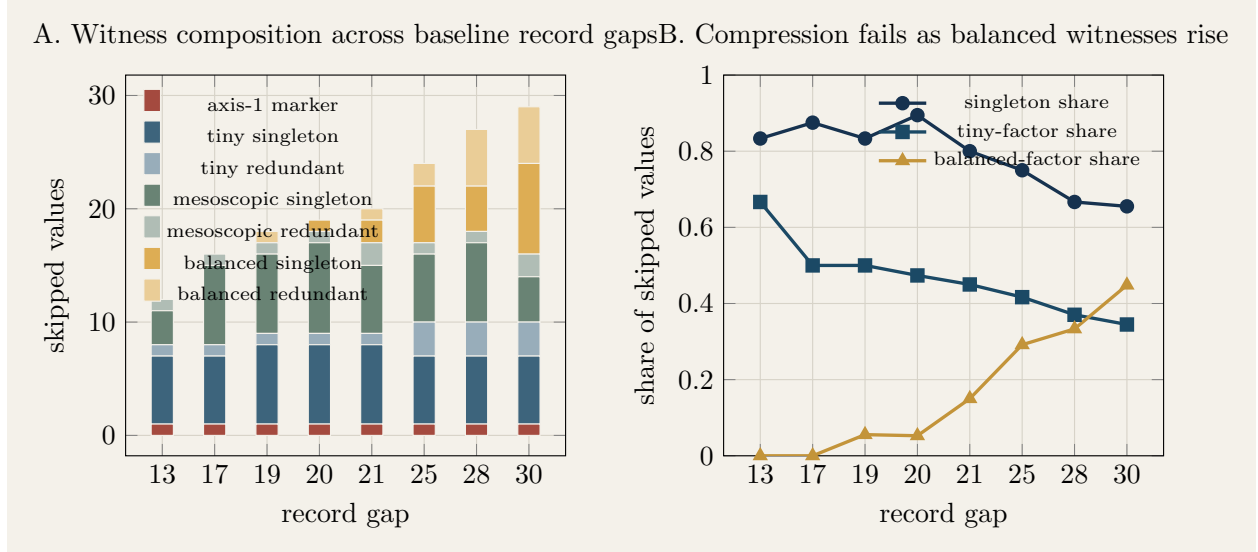


Figure 4: **Witness composition across record gaps.** Left: stacked witness signatures extracted from `results/experiments/run_1000000/record_gap_summary.json`. Every record gap contains exactly one forced axis-1 marker by Corollary 5.6, but the remaining skipped values do not collapse to a fixed small alphabet. Right: three summary shares. The singleton share remains high but not rigid, the tiny-factor share decays, and the balanced-factor share rises sharply. The theorem-level one-point anchor motif survives; an interval-wide compact witness grammar does not.

7.6 The archived variant runs are exact symmetries

The repository stores two nearby update-order runs under `results/experiments/row_immediate_1000000/` and `results/experiments/column_immediate_1000000/`. Section 5 changes their interpretation completely. Proposition 5.3 shows that `row_immediate` is not an empirical perturbation at all; it is mathematically identical to the baseline. Proposition 5.8 shows that `column_immediate` is the exact axis swap. Figure 6 plots these identities directly from the archived sequences. The practical value of the runs is therefore validation, not robustness.

7.7 All first-row gaps through one million steps

For completeness, Figure 7 shows the full empirical distribution of the first-row gaps g_n for $1 \leq n < 10^6$, computed directly from `results/experiments/run_1000000/row_terms.json`. The modal gaps are 4, 3, 5, and 6, but the tail extends to 30. Again, these are exact deterministic frequencies rather than statistical estimates.

8 Discussion

8.1 What is now rigorous

The present paper upgrades several features of the project from observation to theorem.

- The “first two missing values” viewpoint is exact.
- The interleaving $r_n < c_n < r_{n+1}$ is exact.

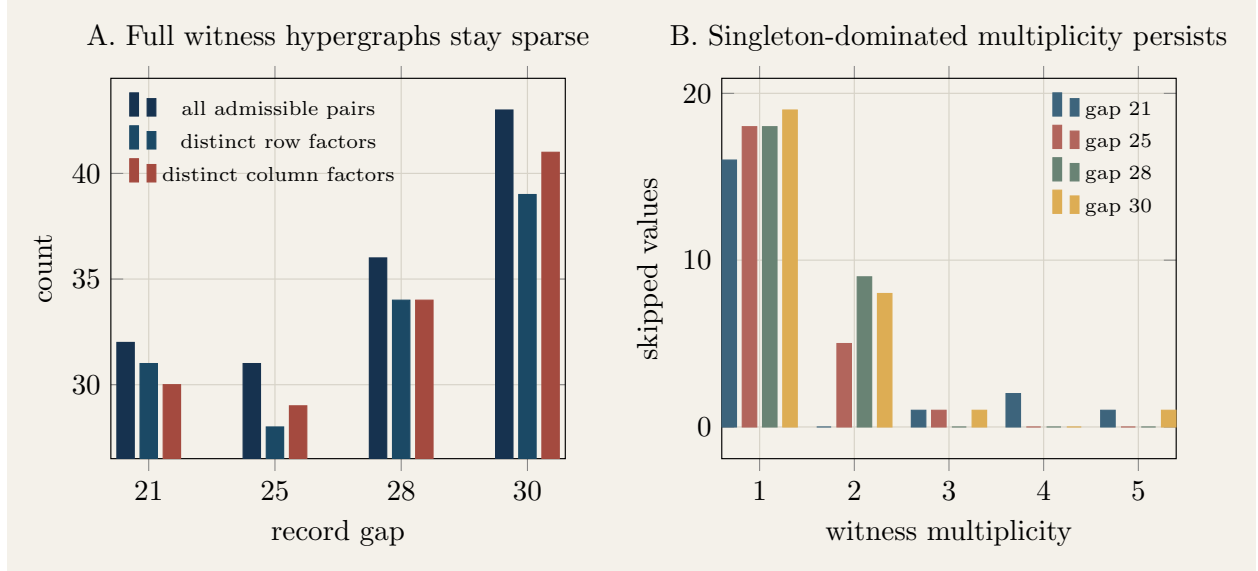


Figure 5: **Selected full witness hypergraphs for late baseline record gaps.** Left: total admissible factor pairs and the numbers of distinct row and column factors used in gaps 21, 25, 28, and 30, from `results/analysis/full_witness_hypergraphs_gap21_25_28_30.json`. Right: multiplicity histograms for the same gaps. Even after full enumeration, the intervals remain singleton-dominated and use nearly as many distinct factors as there are witness pairs. This is evidence against a low-description frontier certificate.

- Each nontrivial row gap contains exactly one column term and therefore exactly one forced axis-1 witness.
- The anchor decomposition expresses the left and right depths of each row gap explicitly in terms of (g_n, δ_n) .
- The two archived update-order runs are exact symmetries of the recurrence.
- Prime separation follows directly from the recurrence.

These statements matter because they separate genuine mathematical structure from implementation artifacts and because they explain several persistent motifs in the data, especially the single anchor marker inside each row gap.

8.2 What the computational evidence now says

The evidence package is also sharper than a simple longer run.

- The baseline generator is reproducible at one million steps and now has a separate checker that agrees through 10^5 steps.
- The late record-gap trajectory reaches 30 in the baseline run and 31 in the axis-swapped run, but those facts remain finite-horizon observations only.
- Large late record gaps can be composite-only, so skipped primes are not the governing phenomenon.

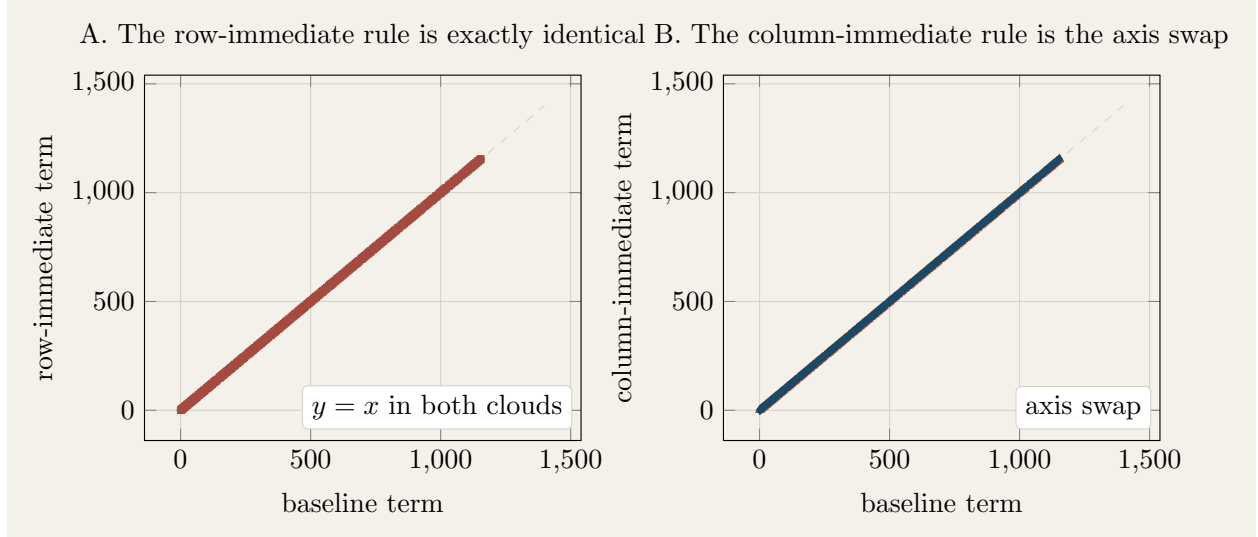


Figure 6: **Exact identities behind the archived update-order runs.** Left: the `row_immediate` sequence pair lies exactly on the diagonal when plotted against the baseline, because Proposition 5.3 proves that the rule is identical to the original recurrence. Right: the `column_immediate` pair lies exactly on the axis-swapped diagonal because Proposition 5.8 proves that it simply exchanges the roles of the two border sequences. These runs are therefore code-level consistency checks, not genuine robustness experiments.

- The anchor geometry is genuine and strongly discriminative on the late matched corpus, yet it fails to stabilize into a bounded correction law.
- The border-prefix and modular mechanisms fail outright.
- The full hypergraph layer yields a descriptive near-miss but not a proof-quality invariant.

This is enough to support a careful negative-result paper. It is not enough to support a theorem-level mechanism claim.

8.3 Remaining limitations

Three limitations are still load-bearing.

- (i) **Finite horizon.** The largest provenly observed baseline record gap is 30, not infinity. No asymptotic claim follows from that alone.
- (ii) **Checker horizon.** The independent checker validates the recurrence through 10^5 steps, not through the full million-step horizon. That is a substantial improvement over the earlier state of the repo, but it is still not a fully independent million-step replication.
- (iii) **Control coverage.** The matched-control and surrogate corpus is real and useful, but it is intentionally bounded: five late record windows, forty-five controls, and fifty surrogates. This is enough to reject several naive mechanism claims, but not enough to prove that every future candidate invariant must fail.

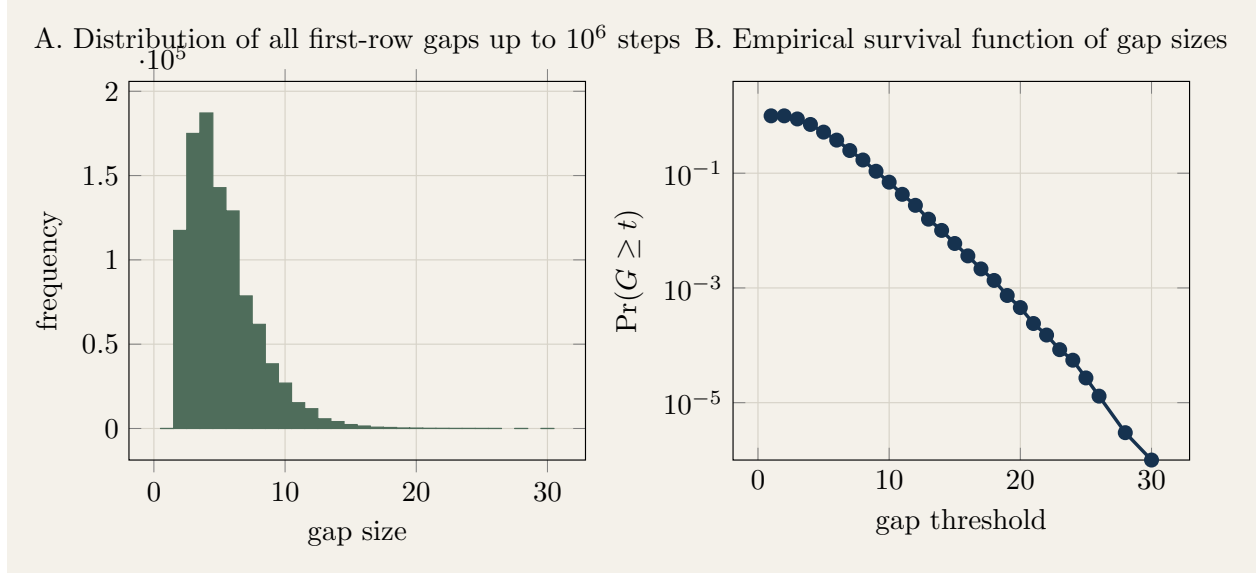


Figure 7: **Distribution of all first-row gaps through the archived million-step run.** Left: exact frequencies of $g_n = r_{n+1} - r_n$ for $1 \leq n < 10^6$. Right: the empirical survival function on a logarithmic vertical scale. The distribution is concentrated on moderate values but has a visible long tail; finite data alone do not distinguish a bounded tail from a slowly moving frontier.

8.4 Open questions

The sharpened picture leaves a more precise list of open questions.

- (1) Is the offset sequence $\delta_n = c_n - r_n$ bounded?
- (2) Can one control the right depth $u_n = r_{n+1} - c_n - 1$ by a horizon-independent invariant extracted from the coupled mex dynamics?
- (3) Is there a full-witness invariant that genuinely uses the mex coupling and survives matched non-record controls, or is every such statistic destined to overlap with generic local product coverage?
- (4) Can one prove arbitrarily long covered frontier intervals in $\mathcal{R}_n \mathcal{C}_n$ by a recurrence-specific argument rather than by generic restricted-product heuristics?
- (5) Conversely, can one construct a rigorous lower-bound mechanism that forces unbounded growth of g_n ?

8.5 Societal implications

This is a pure mathematics and reproducible-computation project. Its direct societal stakes are low. The broader methodological point is that open-problem work around public sequences is especially vulnerable to rhetorical overreach. The repo’s verification layer is valuable precisely because it forces the final paper to distinguish structural theorems, finite-horizon evidence, and failed mechanisms instead of treating all three as the same kind of progress.

9 Conclusion

The boundedness of the first differences of Kimberling’s prime-separator array remains unresolved. What has changed is the precision of the mathematical and computational record. The recurrence has a rigid local geometry: it always chooses the first two missing values of the current product set, its border sequences strictly interleave, every row gap contains exactly one forced axis-1 witness, and the two archived update-order variants are exact symmetries rather than robustness probes. On the experimental side, the baseline million-step corpus extends the observed record-gap trajectory to 30, while the later matched-control audits show that the most natural mechanism candidates fail in specific ways: the anchor/offset lane is descriptive but not bounded-memory, recent-prefix laws are structurally impossible, hypergraph invariants are near-misses rather than proofs, and small-modulus locking is nonpredictive.

The contribution of the paper is therefore narrower than the motivating open problem. It does not settle boundedness, but it does determine what the recurrence guarantees exactly, what the current evidence genuinely supports, and which mechanism stories future work must improve on rather than merely restate.

Table 2: Late matched-control audit summary from the novelty-deepening artifacts. All metrics are deterministic functions of the stored windows.

audit lane	main quantitative outcome	paper-level conclusion
Independent checker	Exact row and column agreement with the baseline through 10^5 steps; checker runtime 0.467 s versus baseline runtime 0.673 s (<code>results/novelty_deepening/checker_agreement.json</code>).	Large-horizon evidence still comes from one generator family, but the update-order logic is no longer unchecked.
Anchor/backbone	Training support of correction pairs is $\{(4, 4), (7, 2), (8, 4)\}$, while the held-out record windows at gaps 28 and 30 require new pairs (17, 1) and (14, 2); bounded-support check is false, though the anchor extremality score achieves held-out AUROC 0.9224 (<code>results/novelty_deepening/item_026_anchor_metrics.json</code>).	The anchor geometry is real and informative, but it fails as a bounded-memory mechanism.
Schedule audit	On 1000 late sampled states, batched and <code>row_immediate</code> schedules agree everywhere, <code>column_immediate</code> matches the axis swap everywhere, and the schedule-independent right-defect depth has AUROC 0.9569 versus 0.5259 for window length alone (<code>results/novelty_deepening/item_027_schedule_metrics.json</code>).	One-step frontier evolution is abelian up to axis swap on the audited states, but the discriminative observable is still just the anchor geometry from Corollary 5.7.
Border-prefix law	The recent-prefix depth needed to cover the five late record windows is 27675, 29372, 92319, 247398, and 729352 respectively (<code>results/novelty_deepening/item_028_prefix_metrics.json</code>).	Any bounded recent-prefix certificate is structurally impossible, because the anchor value c_n requires the ancient row factor 1.
Hypergraph and modular lanes	Full witness cycle rank is 0 on records, controls, and surrogates; the best leaf-fraction near-miss has record-vs-control AUROC 0.8711 but 28.9% matched-control overlap; the best modulus with $q \leq 200$ has AUROC 0.5259 (<code>results/novelty_deepening/item_029_hypergraph_metrics.json</code> , <code>results/novelty_deepening/item_030_modular_metrics.json</code>).	Hypergraph structure yields a descriptive near-miss rather than a credible invariant, and small-modulus locking does not explain the late record windows.

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A Extended Proof Details

A.1 A direct proof that the row gaps always contain a column term

For completeness we isolate the short argument behind Corollary 5.5. Because r_n is the first missing value of $\mathcal{P}_{n-1}$ and c_n is the second missing value of the same set, we have $r_n < c_n$. By Proposition 5.3, once r_n is adjoined, the value c_n is already represented before the next row mex is taken. Therefore $c_n \in \mathcal{P}_n$, which forces $c_n < r_{n+1}$. Hence c_n lies in (r_n, r_{n+1}) . Strict interleaving then excludes all other column terms from the same interval.

A.2 Why Proposition 5.9 is specific to primes

The proof of prime separation uses only one arithmetic fact: if a prime p factors as xy with $x, y \in \mathbb{N}$, then one factor must be 1. This argument does not extend to composites. In fact the whole difficulty of the boundedness question lies in the composite coverage: the late record intervals are frequently composite-only and yet almost fully represented in $\mathcal{R}_n\mathcal{C}_n$.

B Supplementary Experimental Tables

Table 3: Late baseline record gaps and witness summaries from `results/experiments/run_1000000/record_gap_summary.json`.

gap	step	skipped primes	singleton share	tiny share	balanced share	offset before	offset after
19	8475	1	15/18	9/18	1/18	9	1
20	27676	0	17/19	9/19	1/19	13	2
21	29373	0	16/20	9/20	3/20	13	4
25	92320	1	18/24	10/24	7/24	21	4
28	247399	0	18/27	10/27	9/27	11	1
30	729353	0	19/29	10/29	13/29	16	2

Table 4: Repeated million-step runtimes from `results/experiments/runtime_repeats_1000000.json`.

rule	mean runtime (s)	95% CI half-width (s)	largest record gap	record-gap count
baseline	9.944	0.384	30	17
row_immediate	9.975	0.243	30	17
column_immediate	9.628	0.275	31	20

Table 5: Canonical leaf-fraction near-miss from `results/novelty_deepening/item_029_hypergraph_metrics.json`.

quantity	balanced canonicalization	lexicographic canonicalization
record band minimum	0.9474	0.9444
record band maximum	0.9811	0.9808
record-vs-control AUROC	0.8711	0.8156
surrogate failure rate	1.0000	0.9600
matched-control overlap	0.2889	—

C Implementation Details

The minimum artifact set used in the manuscript is:

- `scripts/prime_separator.py`

- `scripts/prime_separator_variants.py`
- `scripts/novelty_deepening.py`
- `scripts/summarize_record_gaps.py`
- `scripts/export_witness_hypergraphs.py`
- `scripts/make_paper_figures.py`
- `results/experiments/run_1000000/contract.json`
- `results/experiments/run_1000000/record_gap_summary.json`
- `results/novelty_deepening/checker_agreement.json`
- `results/novelty_deepening/shared_corpus.json`
- `results/novelty_deepening/item_026_anchor_metrics.json`
- `results/novelty_deepening/item_027_schedule_metrics.json`
- `results/novelty_deepening/item_028_prefix_metrics.json`
- `results/novelty_deepening/item_029_hypergraph_metrics.json`
- `results/novelty_deepening/item_030_modular_metrics.json`
- `figures/figure1_construction.pdf` through `figures/figure7_gap_distribution.pdf`

D Hyperparameter Sensitivity

There are no tunable model hyperparameters in the mathematical recurrence. The only implementation-level parameters are the number of requested steps, the initial size of the extendable coverage array, and the figure-rendering resolution. The mathematical conclusions in this paper are insensitive to the initial array size because the generator dynamically extends the coverage structure as needed. The 600-DPI rendering choice affects only visual sharpness of the PNG exports and not the mathematics.